

REVIEW ARTICLE

Exercise-mediated modulation of autophagy in skeletal muscle

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Although exercise exerts multiple beneficial health effects, it may also damage cellular structures. Damaged elements are continuously degraded and its constituents recycled to produce renovated structures through a process called autophagy, which is essential for the adaptation to training. Autophagy is particularly active in skeletal muscle, where it can be evaluated using specific molecular markers of activation (unc-51-like kinase 1 [ULK1] phosphorylation) and specific proteins indicating increased autophagosome content (increased total LC3, LC3-II, LC3-II/LC3-I ratio). Studies in humans are technically limited but have provided evidence suggesting the activation of autophagy in skeletal muscle through AMP-activated protein kinase (AMPK) and its downstream target ULK1. Autophagy activation is more likely when the intensity is elevated and the exercise performed in the fasted state. The autophagy-gene program and autophagosome content are upregulated after ultraendurance running competitions. However, autophagosome content is reduced after endurance exercise at moderate intensities (50% and 70% of $\text{VO}_{2\text{max}}$) for 60–120 minutes. Autophagosome content is decreased within the first few hours after resistance training. The effects of regular endurance and strength training on basal autophagy remain to be established in humans. One study has reported that acute severe hypoxia increases autophagosome content in human skeletal muscle, which is reverted by 20 minutes of low-intensity exercise. Experiments with transgenic mice have shown that autophagy is necessary for skeletal muscle adaptation to training. Little is known on how genetic factors, environment, nutrition, drugs and diseases may interact with exercise to modulate autophagy at rest and during exercise in humans.

KEYWORDS

endurance training, nutrient intake, protein degradation, protein turnover, resistance training, signaling

1 | INTRODUCTION

Exercise is known to exert multiple beneficial effects and is recommended for the improvement of health and the prevention and treatment of many pathologic conditions.^{1,2} Nevertheless, exercise, as well as nutrient deprivation, hypoxia, infection, high temperature or exposure to toxic chemicals and radiation, may damage cellular structures.³ In all living organisms, damaged elements are continuously

removed and renovated,⁴ facilitating remodeling and adaptation to training in skeletal muscles. In this process, dysfunctional components should be identified, degraded and its constituents recycled as building blocks for the synthesis of new components of improved quality. Autophagy (from the Greek *auto*, “self” and *phagein*, “to eat”) constitutes, in eukaryotic cells, the principal mechanism for the degradation of unneeded components such as damaged organelles, cytoplasmic fractions and protein aggregates.⁵ Autophagy is not

only a critical biological process to avoid disease,⁵⁻⁷ but it is also required for a physiologic adaptation to regular exercise (angiogenesis, mitochondrial biogenesis, insulin sensitivity and muscle hypertrophy).^{8,9,11}

Until now, three main autophagy pathways have been identified. The best-understood mechanism, macroautophagy (hereafter referred to as autophagy), consists of a dynamic process in which a double-membrane cytoplasmic vesicle called autophagosome selectively engulfs dysfunctional or damaged proteins, organelles and fractions of cytoplasm. Subsequently, the completed autophagosome fuses with the lysosome (forming the autolysosome) for an eventual degradation of the sequestered components via lysosomal hydrolases. Chaperone-mediated autophagy utilizes specific chaperones such as hsc70, which identifies and targets cytosolic substrate proteins containing KFERQ-like motifs in their amino acid sequence to be transported to the lysosome for degradation.¹² A third pathway named microautophagy comprises the engulfment of cytoplasmic residues by lysosomal membrane invaginations. The three main autophagy pathways end with lysosome digestion and release of the molecular components of engulfed materials, mostly amino acids, and presumably lipids and sugars, that may follow complete degradation or be recycled by the cell for the *de novo* synthesis of molecules and cellular structures.

An appropriate level of basal autophagy, also referred to as quality control autophagy, is crucial for a correct disposal of cellular and organellar damaged structures to maintain cell homeostasis.¹³ Upregulation or downregulation of autophagy is one of the main mechanisms of cellular adaptation to stress. However, excessive or insufficient autophagy (basal or in response to cellular stress) has been identified as the principal cause of several diseases.¹⁴⁻¹⁷ When cellular stress is increased, for example during hypoxia,¹⁸ limited nutrient availability¹⁹ or exercise,⁹ autophagy flux (defined as a measure of autophagic degradation activity²⁰) is elevated to provide energy substrates and to adapt cellular structures to the newly elevated demands. Skeletal muscle is one of the tissues with highest basal autophagy flux and greater capacity to increase autophagy flux.¹⁹ The maintenance of skeletal muscle health and function requires an adequate level of autophagy and a finely tuned balance between mitophagy (selective removal of mitochondria by autophagy²¹) and mitochondrial biogenesis.^{22,23}

This review examines the current evidence on the influence of exercise on skeletal muscle macroautophagy. Given the limited evidence obtained in humans, some animal findings have also been considered in this review. More specifically, we discuss the effects of exercise modality, duration and intensity on autophagy responses. The influence of regular exercise on skeletal muscle autophagy and the importance of autophagy for skeletal muscle adaptation to training are also addressed.

2 | MOLECULAR MARKERS OF AUTOPHAGIC FLUX IN SKELETAL MUSCLE

At least five different sequential events can be distinguished in autophagy: (a) induction (or activation), (b) formation of the sequestering phagophores (a double-membrane structure), (c) formation of autophagosomes (spheroid organelles still surrounded by a double membrane), (d) fusion of autophagosomes with lysosomes to form autolysosomes (organelles with a single membrane) and (e) hydrolysis of the cargo in the autolysosomes and release of molecular constituents. The term “autophagic flux” is used when referring to the entire process of autophagy (ie from cargo sequestration to its eventual degradation), as opposed to the measurement of the numbers or volume of autophagic elements at any step²⁴ (Figure 1). Currently, there are a considerable number of valid methods to assess autophagic flux in a wide range of living organisms.²⁴ Recent advances have enabled the *in vivo* assessment of autophagic flux in mammalian systems, principally by using autophagy inhibitors and inducers. Nevertheless, this remains unfeasible in humans²⁰ and alternative procedures with limited validity are often applied. For example, a commonly used approach to determine the rate of cargo degradation is the measurement of the rate of general protein breakdown by autophagy. Frequently autophagic flux is indirectly assessed by determining the changes in molecular markers of the different steps involved in autophagy, mainly adenosine monophosphate-activated kinase (AMPK), the unc-51-like kinase (ULK), the Forkhead Box (Fox) O (FoxO), the microtubule-associated protein 1A/1B-light chain 3 (Atg8/LC3) and the sequestosome 1 (p62/SQSTM1) (Figure 2). This approach has limitations since certain stimuli may result in changes of specific molecular markers of autophagy, unrelated to real modifications in autophagic flux.²⁵

AMP-activated protein kinase (AMPK), which plays a critical role in energy homeostasis,²⁶⁻²⁸ is also considered a crucial activator of autophagy²⁴ and mitophagy.²⁹ AMPK is activated by phosphorylation of the 172-threonine residue of the α chain when the AMP/ATP ratio is increased.³⁰ Exercise-elicited AMPK α Thr¹⁷² phosphorylation depends on the training status and the characteristics of the exercise.^{31,32} Animal studies indicate that AMPK activation is necessary for exercise to induce autophagy^{33,34} while myristoylation of AMPK β is needed to induce mitophagy.²⁹ AMPK promotes autophagy by directly phosphorylating and activating ULK. ULK activation is essential for the initiation of autophagosome formation.^{35,36} Among the five isoforms of ULK, ULK1 is considered the predominant isoform involved in autophagy.²⁴ ULK1 together with the autophagy-related 13 (Atg 13) and the focal adhesion kinase family-interacting protein of 200 kDa (FIP200) form an indispensable complex

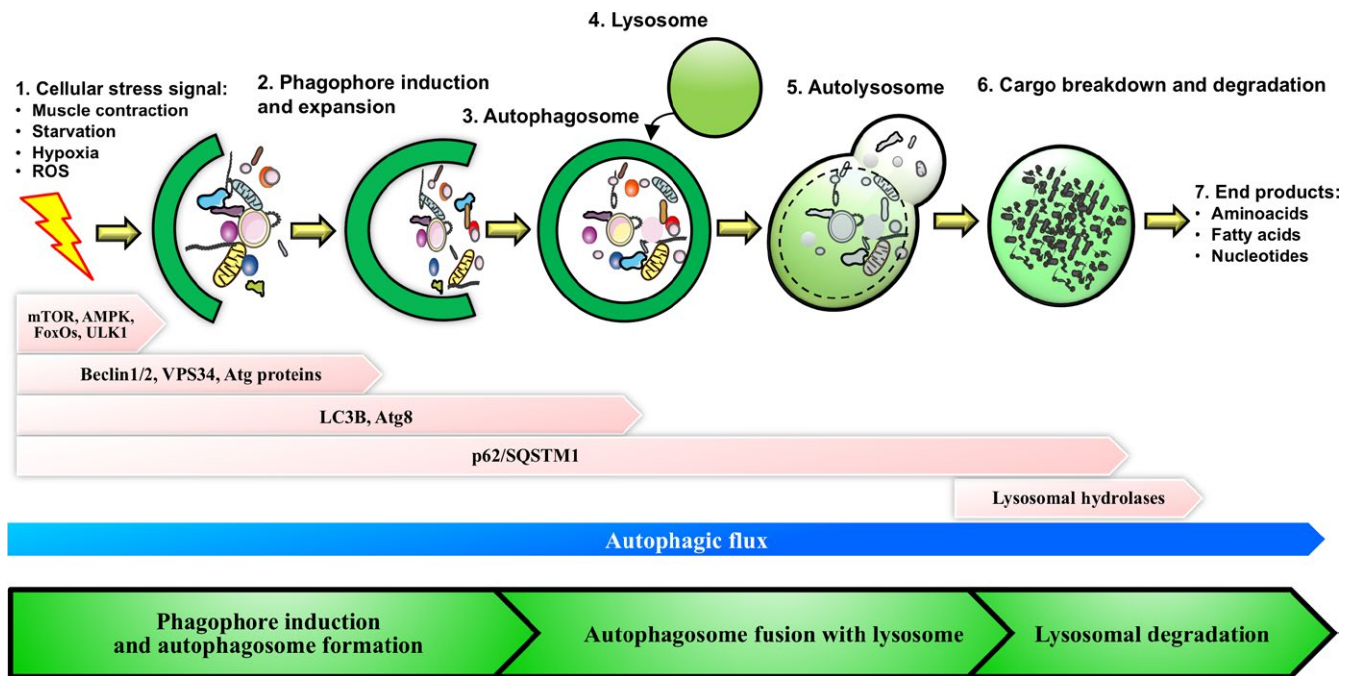


FIGURE 1 Schematic representation of the steps involved in autophagy. Autophagy begins with the induction and nucleation of the phagophore (isolation membrane) from the initiation complex. Phagophore undergoes expansion and closes on itself forming the autophagosome while enclosing cellular components. The completed autophagosome fuses with lysosomes to form an autolysosome. Finally, internal cellular components of the autolysosome are degraded alongside its inner membrane by lysosomal hydrolases into the final end-products (essentially amino acids, fatty acids and nucleotides), which can be recycled

for autophagy induction.^{35,37} Although the exact AMPK-mediated ULK1 phosphorylation site(s) remains unclear Ser³¹⁷,⁴⁶⁷,⁵⁵⁵,⁵⁷⁴,⁶³⁷ and⁷⁷⁷, as well as Thr⁵⁷⁴ have been implicated.^{24,38,39} Nevertheless, it has also been reported that Ser⁵⁵⁵ phosphorylation of ULK1 is not sufficient to increase autophagosome content in rodent muscle and human myotubes.⁴⁰

In contrast, phosphorylation of ULK1 at the 757-serine residue by the mammalian target of rapamycin complex 1 (mTORC1) prevents ULK1 activation.³⁹ AMPK inhibits mTORC1 complex by direct phosphorylation^{41,42} and also by phosphorylating and activating Tuberous Sclerosis Complex 2 (TSC2).⁴³ Thus, AMPK acts simultaneously on two processes that synergize to activate autophagy: directly, by activating ULK1, and indirectly by impeding mTOR-dependent inhibition of ULK1.

When activated, ULK1 phosphorylates and activates the protein beclin-1 (encoded by gene *BECN1*) on Ser¹⁴ (mice nomenclature; Ser¹⁵ in humans), which seems necessary for complete activation of autophagy.³⁶ Nucleation and expansion require of a fine functioning of a complex formed by beclin-1, vacuole protein sortin 34 (Vps34) and 15 (Vps15), as well as activating molecule in BECN1 regulated autophagy protein 1 (AMBRA1), which interacts with the aforementioned ULK1/Atg13/FIP200 complex.⁴⁴ B-cell lymphoma-2 (BCL2) is an anti-apoptotic and anti-autophagy protein that inhibits autophagy by binding to the BH3 domain of the

autophagy protein beclin-1 at the endoplasmic reticulum.⁴⁵ By binding to beclin-1, BCL2 blocks beclin-1 binding to Vps34 impeding autophagy initiation.^{46,47} Induction of autophagy requires the disruption of the BCL2-beclin-1 complex. The important role played by ULK1 in autophagy has been confirmed using loss-of-function genetic models showing that lack of ULK1 impedes the conversion of LC3-I to LC3-II in mice.³⁹

ULK1 effects may be reinforced by FoxOs (FoxO1, FoxO3 and FoxO4), among which FoxO3 upregulates autophagy-genes expression in skeletal muscle.⁴⁸ FoxO3 activation is facilitated in catabolic conditions such as ultraendurance exercise^{50,51} and by nutrient deprivation and lack of growth factors (IGF-1, insulin, etc.), resulting in reduced intracellular Akt (also known as protein kinase B, PKB) activity.⁵² However, during sprint,⁵³ endurance⁵⁴ and resistance exercise,⁵⁴ Akt is phosphorylated and activated, which may negatively regulate autophagy by phosphorylating threonine residues of FoxO3.^{48,51} PhosphoThr³²-FoxO3 remains in the cytosol⁵⁵ and only translocates to the nucleus to stimulate the autophagy-gene program when dephosphorylated.^{56,57} Akt can also inhibit autophagy by blunting Thr¹⁷²-AMPK phosphorylation via Ser⁴⁸⁵-AMPK α 1/Ser⁴⁹¹-AMPK α 2 phosphorylation.^{30,53,58,59} The exercise-induced activation of Akt is more prominent when the exercise is performed in the fed than the fasted state, likely due to the additive effect of the postprandial increase in insulin and exercise on Akt

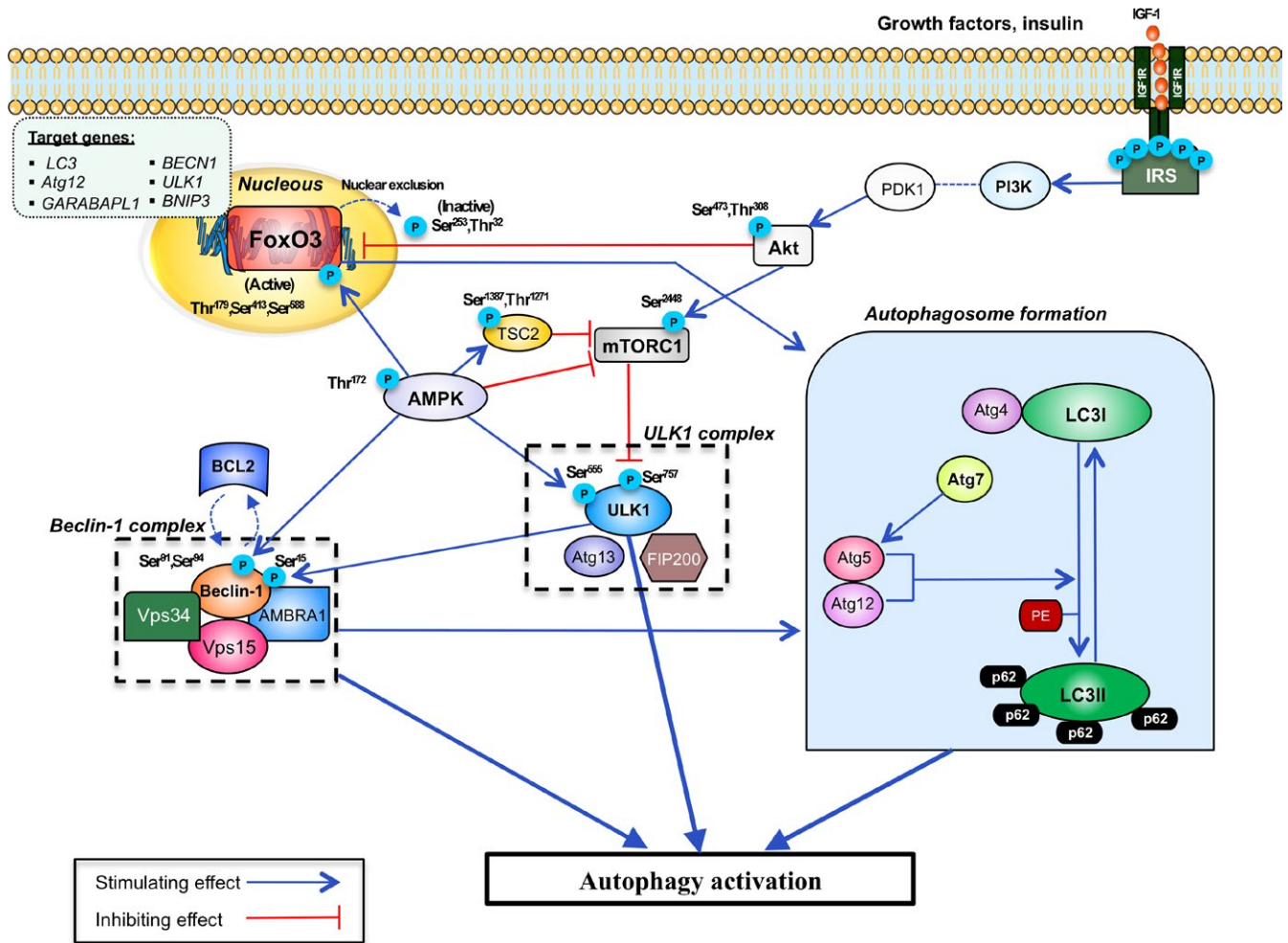


FIGURE 2 Signaling pathways regulating autophagy activation in skeletal muscle. Exercise-induced stress in muscle fibers activate a complex cascade of molecular events initially driven by FoxO3, AMPK and mTOR, depending upon the nutrient state and the type of exercise (modality, intensity and duration). AMPK induces autophagy activation by directly phosphorylating ULK1. Anabolic stimuli activate insulin signaling leading to mTOR activation, which acts as an autophagy inhibitor via downregulation of the ULK1-complex. AMPK blocks mTOR directly or by activating TSC2 via phosphorylation. FoxO3 coordinates a transcriptional program that promotes several autophagy genes. FoxO3 is phosphorylated and inhibited by Akt. Phosphorylated FoxO3 by Akt leads to exclusion from the nucleus. FoxO3 is upregulated by AMPK during prolonged energy deficit. Phagophore formation and expansion are controlled by the fine integration of proteins present in the beclin-1 complex. Exercise-mediated BCL2 phosphorylation induces beclin-1-to-BCL2 dissociation permitting autophagosome nucleation. Nucleation and expansion of the autophagosome requires LC3-I lipidation to LC3-II mediated by various conjugation reactions by Atg proteins. Sequestration of target cellular components for degradation is accomplished by ubiquitin, which selectively recruits and binds to p62. Recruited p62 directs the targeted material to be degraded within the growing autophagosome by binding to LC3-II on the autophagosomal membrane

activation.⁵³ Moreover, AMPK phosphorylates FoxO3a in Ser⁵⁸⁸ increasing its nuclear localization and activity.^{60,61}

Prolonged endurance exercise is likely needed to promote FoxO3 signaling^{49,50} although few hours after a single bout of bicycling for 120 minutes at 60% of VO₂max, *FoxO3* mRNA and protein levels were already increased.⁶² In contrast to endurance exercise, FoxO3 protein levels remained unchanged after a single bout of resistance exercise.⁶² The increase in FoxO3 protein after endurance exercise is also observed in the trained state.⁶² The importance of FoxO3 for autophagy is further supported by suppression of autophagic flux in *FoxO3* knockout mice.⁵⁶

The ubiquitin-like protein Atg8/LC3 can be found in phagophores, autophagosomes and, to a lesser extent, in autolysosomes. Native LC3, or proLC3, is proteolytically cleaved by autophagy-related 4 (Atg4) (a protease) releasing LC3-I, to which phosphatidylethanolamine is conjugated to generate the lipidated form of LC3 called LC3-II. LC3-II is the only protein marker that is reliably associated with completed autophagosomes (although it can also be found in phagophores)²⁴ and its levels correlate with autophagosome number.⁶³ During autophagy, there is an increased conversion of LC3-I to LC3-II and the ratio LC3-II/LC3-I is augmented. In the latest step of autophagy, the sequestosome

1 (p62/SQSTM1), an acceptor for ubiquitinated substrates, is reduced due to autolysosomal degradation.⁶⁴ Conversely, the accumulation of p62/SQSTM1 has been interpreted as a marker of autophagy inhibition.^{35,65} Nevertheless, p62 changes should be interpreted cautiously because the level of p62 may also change by mechanisms unrelated to autophagy.^{24,66} Moreover, using either the content of LC3-II or the ratio LC3-II/LC3-I as a marker of autophagic flux lacks specificity, as the content of autophagosomes depends on both the rate of formation and the rate of lysosomal degradation.^{63,66} This is further complicated by the fact that the conversion of LC3-I to LC3-II is cell-type-specific and dependent on the stimulus used to induce autophagy.²⁴

Due to these limitations, other methods including transmission electron microscopy and fluorescent techniques are used to assess autophagic flux in animals or cells.²⁴ In humans, however, it is not possible to employ molecular reagents, and the use of transmission electron microscopy should be limited to few measurements, which require repeated muscle biopsies. The assessment of mRNA encoding for autophagy proteins in combination with immunohistochemical techniques may help to obtain a more valid assessment of autophagy in human skeletal muscle.⁶⁷

3 | ENDURANCE EXERCISE AND AUTOPHAGY IN SKELETAL MUSCLE

Salminen and Vihko⁶⁸ were the first reporting an augmented number of autophagic vacuoles following strenuous endurance exercise in rat skeletal muscle. Nonetheless, previous work had linked exercise to activation of autophagy in other tissues.⁶⁹ Since then, acute endurance exercise has been shown to activate autophagy at the gene and protein level in rodent skeletal muscle,^{8,50,70-73} cardiac muscle,⁷⁴ as well as in other tissues⁹ despite some controversy.⁷⁵ In humans, limited and controversial results have been reported to date.^{40,50,67,76,77}

Jamart et al.⁴⁹ reported upregulation of autophagy following a 28-hours ultraendurance running (200 km), but this conclusion was based only on mRNA analysis. Muscle biopsies obtained after a 150 km running competition (18 hours) revealed a fivefold increase in LC3-II, indicating increased autophagosome content.⁵⁰ In contrast, a reduction of LC3-II has been observed after less strenuous endurance exercise protocols,^{40,67,76} whereas LC3-II did not change after 30 minutes cycling at 70% of VO_2max ³⁴ or 15 cycling at 40% of VO_2max with blood flow restriction.³⁴ Moller et al.⁷⁶ studied eight young recreational athletes before, and immediately after bicycling at 50% of VO_2max for 60 minutes, and 30 minutes after the end of exercise, on two occasions: (a) close to the end of a 36-hours fast, and (b) during continuous

glucose infusion at $0.2 \text{ kg}\cdot\text{h}^{-1}$. Autophagy signaling through ULK1 was increased 30 minutes after the end of the exercise, while LC3-II and the LC3-II/LC3-I ratio were reduced at the end of exercise and remained at this level 30 minutes later (p62 was unchanged). Therefore, Moller et al.⁷⁶ study indicates a potential activation of autophagy, accompanied by a reduction of autophagosome content at the end of exercise, regardless of the nutrient background.

Schwalm et al.⁶⁷ studied the autophagy response to an acute bout of endurance exercise in well-trained athletes (minimal VO_2max of $50 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$). The volunteers were studied under fasted and fed conditions and were divided into three groups: control ($n=8$), low-intensity (LI, $n=8$) and high-intensity (HI, $n=7$). Both groups cycled for 2 hours, the LI group at 55% and the HI at 70% of VO_2peak . As in Moller et al.,⁷⁶ ULK1 was phosphorylated at site Ser³¹⁷ (mice nomenclature). This effect was greater when the high-intensity exercise was performed in the fasted state. In both the fed and the fasted states, LC3-II protein amount and LC3-II/LC3-I ratio were reduced after LI and HI, indicating decreased autophagosome content.⁶⁷ The autophagy transcriptional program was also activated, as shown by the increased LC3B, p62, GABA (A) receptor-associated protein like 1 (GABARAPL1), and Cathepsin L mRNAs observed after HI but not after LI.⁶⁷ Interestingly, Schwalm et al.⁶⁷ study provide some evidence indicating that prolonged endurance exercise may be a more potent stimulus for autophagy than a short fasting period.

More recently, Fritzen et al.⁴⁰ studied skeletal muscle markers of autophagy after 70 minutes of one-leg knee extension exercise. Compared to the resting leg, the exercised leg showed decreased LC3-II levels and LC3-II/LC3-I ratio suggesting reduced autophagosome content (p62 protein levels remained unchanged).⁴⁰

Masschelein et al.¹⁸ suggest that the effect of exercise on autophagy may vary depending on the prevalent autophagy flux background. Three-hour exposure to severe hypoxia ($\text{F}_1\text{O}_2=0.107$) increased autophagosome content in 11 young MZ twins, as reflected by a 45% increase in LC3-II and a 29% increase in the LC3-II/LC3-I ratio in hypoxia compared with normoxia, accompanied by a 25% reduction of p62 protein level.¹⁸ After the 3 hours in hypoxia, subjects performed a 20-minutes exercise bout on a cycle ergometer at $81.4\pm 3.2\%$ of VO_2max in hypoxia, which was compared to a control exercise performed on a previous day at the same absolute intensity but in normoxia (representing $50.7\pm 2.3\%$ of VO_2max in normoxia). The LC3-II and LC3-II/LC3-I ratio changes induced by the 3-hour passive exposure to hypoxia were reversed by exercise. However, p62 remained 15% lower than in normoxia, after exercise in hypoxia.

In contrast to human studies, most rodent experiments indicate that autophagy is stimulated during endurance exercise.^{8,70,72,78} He et al.⁸ convincingly demonstrated that

autophagy is already induced 30 minutes after the start of a 110 minutes long treadmill running session at 75% of their maximal running capacity. These authors used transgenic mice expressing a green fluorescent protein (GFP)-labeled marker of autophagosomes (GFP-LC3). The number of autophagosomes in skeletal and cardiac muscle was increased after 30 minutes and continued to rise until 80 minutes, plateauing thereafter (Figure 3). This was a generalized response observed in several muscle groups including predominantly slow and fast exercised muscles. These findings were supported by biochemical evidence as reflected by the increase in LC3-II and conversion of the non-lipidated form of LC3-I to LC3-II, and degradation of the autophagy substrate protein p62. Thus, this study provides validity, albeit in a rodent model, for the use of these molecular markers to assess changes in autophagosome number and flux in skeletal muscle indirectly. Interestingly, He et al.⁸ also reported exercise-induced autophagy in islet β -cells and liver adipose tissue. Biochemical evidence (based on increased LC3-II) also suggested exercise-induced autophagy in adipose tissue. Moreover, this study showed that exercise-induced autophagy requires disruption of the BCL2-beclin-1 complex through a molecular mechanism that remains unknown.⁸

Some rodent studies indicate that a physiologic autophagy response to exercise is required to maintain a normal carbohydrate metabolism and skeletal muscle insulin sensitivity during exercise,⁸ but this has not been confirmed by others using muscle-specific knockout mice.⁷⁹

The underlying mechanisms for discrepancies between rodent and human experiments remain unknown. It has been speculated that rodents may use macroautophagy to release energy substrates while this mechanism is prevented in humans to favor the use of alternative metabolic pathways.⁴⁰

Although unlikely, a rise of autophagosome flux accompanied by a greater increase in autophagosome degradation, resulting in reduced autophagosome content, cannot be ruled out in human skeletal muscle after endurance exercise.

4 | EXERCISE PERFORMANCE AND AUTOPHAGY

Rodent loss-of-function genetic models have shown that defective autophagy may hamper exercise performance mainly due to pathologic alterations in skeletal muscle. These include loss of muscle force, reduced specific tension, increased oxidative stress, pathologic changes in mitochondrial structure and function, reduced capacity to cope with eccentric exercise-induced muscle damage, and defective adaptation to exercise.^{23,48,80} Nonetheless, non-pathologic outcomes are seen in other tissues with autophagy deficiency, suggesting that skeletal muscle, compared to other tissues, may be particularly sensitive to defective autophagy.⁶⁵

Maximal exercise performance is reduced in *BCL2*^{AAA} mice and in transgenic mice having decreased beclin-1 protein expression in skeletal muscle.⁸ In contrast, no impairment of maximal exercise performance was observed in sedentary *Atg6*^{+/-} mice, which are haplodeficient for the autophagy-related 6 (*Atg6/BECN1*), a class III PI3K necessary for autophagosome formation, compared to sedentary WT animals. Nonetheless, *Atg6*^{+/-} mice fail to improve their endurance performance after 4 weeks of voluntary wheel running when compared to WT animals.¹⁰ This lack of improvement in endurance was accompanied by a blunted adaptation to endurance training.

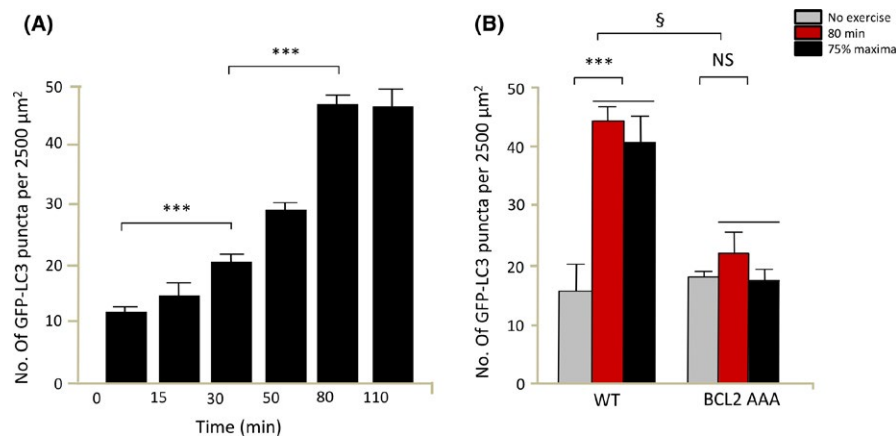


FIGURE 3 (A) Quantification of skeletal muscle (vastus lateralis) data from mice transgenically expressing a green fluorescent protein (GFP)-labeled marker of autophagosomes at serial time points after exercise (mean $\pm$ SEM of 10 tissue sections) $^{**}P < .01$, $^{***}P < .001$. One-way Anova. Modified from He et al.⁸ (B) Quantification of skeletal muscle data from GFP-LC3 transgenic mice (*BCL2* AAA) or wild type before exercise, and following 80 min of exercise or 75% of maximal running capacity. $^{*}P < .05$, $^{**}P < .01$, $^{***}P < .001$, one-way ANOVA to compare between groups; $^{§}P < .001$, two-way ANOVA to compare the magnitude of changes between different groups in mice of divergent genotypes; NS, not significant. Modified from He et al.⁸

Muscle-specific deletion of autophagy-related 7 (*Atg7*), which impedes autophagosome formation, resulted in a loss of muscle mass and specific tension (force/cross-sectional area), which was exacerbated with ageing, and was accompanied by morphologic features of myopathy.²³ Moreover, *Atg7* knockout mice experience greater muscle damage with eccentric muscle contractions.⁷⁹ Similarly, peroxisome proliferator-activated receptor γ coactivator 1- α (*Pgc-1 α*) knockout mice have reduced mitochondrial volume, lower mitochondrial turnover due to decreased mitophagy and mitochondrial biogenesis, and reduced exercise capacity.⁷³

5 | INFLUENCE OF TRAINING ON BASAL AUTOPHAGY

Rodent experiments have shown that predominantly oxidative muscles have greater basal autophagy, and expression of autophagy and mitophagy proteins than predominantly glycolytic muscles.^{10,21} Also in rodents, Lira et al.¹⁰ showed that 4 weeks of voluntary wheel running led to a significantly increased basal autophagy (measured at least 28 hours after the last training session) in the *plantaris* muscle (mixed-fiber type). Although autophagy protein expression was increased, no significant changes were observed in basal autophagy and mitophagy protein content in the soleus muscle (predominantly oxidative).¹⁰ The increased basal autophagy elicited by training may be in part mediated by *Pgc-1 α* , since muscle-specific overexpression of *Pgc-1 α* results in increased basal autophagy flux in mixed-fiber muscles, like *plantaris* in mice.¹⁰ *Pgc-1 α* overexpression upregulates the mitophagy protein BCL2/adenovirus E1B 19 kDa interacting protein 3 (Bnip3), although not at the transcriptional level, to counteract the increased mitochondrial biogenesis, because a mismatch between mitochondrial biogenesis and mitophagy would result in accumulation of dysfunctional mitochondria. The mechanism by which *Pgc-1 α* overexpression upregulates autophagy flux in skeletal muscle remains to be elucidated.

6 | RESISTANCE TRAINING AND AUTOPHAGY IN SKELETAL MUSCLE

Only two studies have determined the effects of a resistance training session on autophagosome content in human skeletal muscle focusing on the first hours after the end of the resistance exercise bout^{81,82} while the effects of resistance training on basal autophagy in humans have not been explored. This is in contrast with the reported reduced autophagosome content after resistance training in rodents.⁸³

In young and older adults, autophagosome content was reduced 3, 6 and 24 hours after a single bout of resistance exercise consisting of 8 sets of 10 repetitions at 70% one-repetition maximum with 3 minutes of rest in between each set, but the fractional breakdown rate of muscle proteins remained unaltered because protein degradation via the ubiquitin proteasome system (UPS) was increased.⁸¹ The same research team examined the effects of a supplement containing essential amino acids (EAA; 0.35 g·kg⁻¹ Lean Mass⁻¹) and two different amounts of carbohydrate (CHO; one group 0.5 g·kg Lean Mass⁻¹ and another group 1.4 g·kg Lean Mass⁻¹) on muscle protein synthesis and degradation in 12 subjects divided into two groups (n=6). Subjects received the supplement 1 hour after 10 sets of 10 repetitions of bilateral leg extension at 70% of their predetermined one-repetition maximum.⁸² The biopsies obtained 1 hour after the ingestion of the supplements showed a significant reduction of LC3-II without significant changes in LC3-I, suggestive of reduced autophagosome content, without significant differences between groups.

7 | REDUCED BASAL AUTOPHAGY IN SKELETAL MUSCLE CONTRIBUTES TO SARCOPENIA BUT REGULAR EXERCISE ANTAGONIZES THIS EFFECT

It has been postulated that the loss of muscle mass and force with ageing (sarcopenia) may be caused by dysregulation of mitochondrial function, dynamics (fission/fusion) and turnover (mitophagy/biogenesis).⁸⁴ Regular exercise is the best treatment available to slowdown the development of sarcopenia with ageing,⁸⁵ improving mitochondrial quality and perhaps the number of quiescence satellite cells, a process that depends on basal autophagy.⁸⁶ Ageing is associated with reduced mitofusin 2 (MFN2), which in turn decreases autophagic flux in skeletal muscle.⁸⁷ Regular exercise increases MFN2 expression and by this way may normalize autophagy attenuating the stimulus for sarcopenia, according to the mitochondrial theory of sarcopenia.⁸⁸ Nevertheless, experiments in humans are required to determine the effects of exercise in basal skeletal muscle autophagy in elderly.

8 | CONCLUDING REMARKS AND FUTURE PERSPECTIVES

Healthy adaptation to exercise training is only possible with a proper balance between the mechanisms for removal and recycling of damaged or dysfunctional cellular components and the formation of new cellular structures in skeletal muscles.^{22,23} Future studies are needed to determine how

different exercise modalities may modulate autophagy in humans depending on sex, age, health status and training background. Little is known about the environmental effects on autophagy. Nothing is known about the interaction between the genotype and the exercise-induced autophagy in humans. Future studies should also examine how drugs which may modify autophagy, like metformin, could alter the adaptations to exercise. Thus, a lot remains to be discovered in this field.

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DISCLOSURE

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